Do Standard Neural Networks Learn Local Block-Spin Renormalization Group Maps?

A Controlled Negative Result on the 2D Ising Majority-Vote Task

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Abstract: We investigate whether standard neural networks (MLP with GELU activations) exhibit *RG-like advantages* on the 2D Ising majority-vote block-spin map, a deliberately narrow local benchmark whose target is well approximated by a linear-threshold rule. The main supported evidence in this revision is a unified 10-seed same-scale benchmark. At the critical point β_c and lattice size $L = 16$, the MLP achieves test MSE 0.62 ± 0.65, while a single linear layer achieves 0.17 ± 0.09 on the same task; the linear model's lower mean error is clear, but the strength of the difference is test-sensitive because the MLP variance is large (Welch $p = 0.055$, Mann-Whitney $p = 0.026$, Cohen $d = -0.98$). Some off-critical same-scale settings show an even stronger linear advantage, e.g. $d = -8.36$ at $(L, \beta) = (16, 0.30)$ where both models have small seed variance. These results support a scoped negative claim: on this local majority-vote task, extra fully connected non-linearity does not yield an RG-like advantage over a linear baseline. We additionally ran a true whole-lattice tiled transfer experiment under a reduced exploratory budget; it is methodologically informative but not decisive, and it does not exhibit a stable monotone scale-degradation law. Dedicated sampler diagnostics further show strong critical slowing down, with $\tau_{\text{int}}(|M|) \approx 37$ sweeps at $(L, \beta) = (16, \beta_c)$, so all cross-scale claims are treated as exploratory rather than decisive evidence for RG-like behavior. A translation-equivariant CNN reaches near-zero MSE in most tested settings, but not all, showing that locality and translation structure are sufficient to learn the supervised block-spin map, but are not by themselves evidence of RG-flow learning. Random-label pilots and a compressed turbulence feasibility note are retained only in the appendix.

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0.1 Introduction

The renormalization group (RG), developed by Wilson [43] and crystallized in Kadanoff's block-spin picture [18], describes how physical systems behave across scales. At a critical point, the RG transformation converges to a fixed point; the approach to this fixed point is governed by critical exponents [12, 46].

The block-spin picture provides an intuitive mechanical picture of RG. Imagine a lattice of spins, each pointing either up (+1) or down (-1). The RG transformation groups nearby spins into blocks — for a 2×2 block size Λ, four adjacent spins become one coarse-grained spin. To determine the new spin, we take a majority vote: if three or four spins in the block point up, the coarse-grained spin points up; if three or four point down, the coarse-grained spin points down. This reduces an $L \times L$ lattice to an $L/2 \times L/2$ lattice. Repeating this operation at multiple scales generates the RG flow. At each step the system is described by fewer degrees of freedom but should represent the same physics at longer wavelengths.

At the *critical temperature* β_c, something remarkable happens: fluctuations occur at *all* scales simultaneously. A spin at the critical point is correlated with other spins at all distances — correlation functions decay as a power law, not exponentially. There is no characteristic length scale in the system (the correlation length ξ diverges to infinity). This is why the critical point is where RG fixed points live: at β_c, the system looks statistically the same no matter what magnification you use. Zoom in on a critical system and you see the same pattern of large and small fluctuations you saw before zooming.

The "fixed point" in RG language is a system that is *invariant* under the coarse-graining transformation. If you apply the block-spin transformation to a system already at the fixed point, you get back a system with the same statistical properties — same correlation functions up to a rescaling of length. The fixed point is the intersection of all RG flows starting from different microscopic parameters (different temperatures, fields, couplings). Physical quantities near the fixed point obey power laws; the exponents of these power laws are the critical exponents. They are universal: they depend only on the dimensionality and symmetry of the order parameter, not on the microscopic lattice details.

The question of whether neural networks implement RG-like computations is therefore a question about whether they discover, through training alone, this hierarchical coarse-graining procedure. If a standard architecture trained on physical data implicitly learns to block-average across scales, this would constitute a strong inductive bias toward the kind of scale-invariant reasoning that RG formalizes.

The hypothesis that neural networks might *implement* RG-like computations was proposed by [26], who observed that restricted Boltzmann machines learn effective couplings matching RG flow in the 2D Ising model. Subsequent work formalized this idea through information-theoretic and explicit coarse-graining constructions [20, 22, 30, 16, 2].

We take a different stance: can *standard* architectures (MLP with GELU, no explicit RG structure) learn block-spin transformations, and do they show quantifiable signatures of RG-like behavior? Our emphasis is deliberately narrow: the target is the local majority-vote block-spin map, not a general non-linear RG flow. This question still matters for the interpretability of learned coarse-graining and surrogate models, but any conclusion must be read at that same scoped level.

The broader significance extends beyond the Ising model. If standard neural networks already implement RG-like computations, this would partially explain their striking success in modeling multi-scale physical systems such as turbulent flows [32, 10], crack

propagation in heterogeneous materials [11, 34], and material defect dynamics. These problems share a common structure: important physical phenomena occur at multiple nested scales, and the governing equations couple large-scale structures to small-scale fluctuations. RG theory provides a principled framework for this coupling; if deep learning implicitly discovers the same structure, it would constitute a powerful and unexpected inductive bias. Conversely, if standard architectures do *not* exhibit RG-like behavior, this raises the fundamental question of what *other* inductive biases they do rely upon — and whether those biases are sufficient for robust extrapolation in physical science. This connects to the broader research program of "machine learning meets theoretical physics" [26, 30, 22, 20], which seeks to understand the relationship between the structure of physical law and the structure of learned representations.

Research Questions

1. Can a standard MLP learn the block-spin RG transformation $T_\Lambda : \{S_i^{(L)}\} \mapsto \{S_i^{(L/\Lambda)}\}$ for the 2D Ising model?

   Here, "learn" means *supervised regression*: we generate examples of fine-grained spin configurations S^{fine} along with their coarse-grained labels S^{coarse} computed via the majority-vote block-spin transformation (Eq 3), and train the network to predict S^{coarse} from S^{fine}. This is not an unsupervised flow-based model nor a generative model; it is a direct function approximation task. The question is whether a standard MLP architecture can represent the block-spin map without having it hardwired into the architecture.

2. Is this learning statistically meaningful (vs. random baselines)?

   Any sufficiently expressive model can memorize a finite training set. Our original design therefore included a random-label sanity check. In the present revision that control remains appendix-level because it was produced under a separate 3-seed pilot protocol and is not part of the benchmark of record.

3. Does the learning quality degrade with scale distance, as RG theory predicts?

   RG theory's central quantitative prediction is that information about the microscopic scale is progressively lost under repeated coarse-graining. If a network is trained on block-spin targets at one lattice size L and then tested at a different size L', the test error should increase monotonically with $|L - L'|$, because the RG flow has moved further away from the training scale. This cross-scale transfer is *the* defining operational test of RG-like behavior in our framework. It is qualitatively distinct from within-scale learning (which merely confirms the network can represent the block-spin map at a single scale) and requires the network to have discovered the *hierarchical* structure of scale-coupling, not just a local majority-vote rule. In the present revision we therefore treat monotone degradation as a hypothesis whose operationalization remains unstable for the current local-rule benchmark, rather than as a settled diagnostic.

4. Does the framework extend to a neural turbulence closure reproducing Kolmogorov spectrum scaling?

   Kolmogorov's 1941 hypothesis [21] for fully developed turbulence states that the energy spectrum in the inertial range follows $E(k) \propto k^{-5/3}$, where k is the wavenumber.

This power law emerges from a cascade: large eddies inject energy at the forcing scale, break into smaller eddies, and so on until viscosity dissipates at the Kolmogorov microscale. The $-5/3$ exponent is remarkable because it can be derived almost entirely from dimensional analysis plus the assumption of universality (the inertial range statistics depend only on the energy cascade rate ϵ). If a neural network trained *only* on subgrid stress closures (the small-scale fluctuations unresolved by the simulation grid) can reproduce this exponent, it would constitute strong evidence that the network has learned an RG-like scale-coupling in a physically non-trivial setting beyond the controlled Ising testbed. This would be striking because the network never sees the inertial range directly during training; the correct exponent would have to emerge from the physics embedded in the training data.

Operational Targets and Hypotheses

We retain the labels R1–R3 to organize the paper, but in this revision they do not all carry the same evidentiary status. R1 motivates the original random-label sanity check, R2 is best read as a *transfer hypothesis* to be tested under a unified multi-scale protocol, and R3 is an exploratory physics-extension target.

Concretely:

(R1) **Appendix-level sanity check:** Test MSE on block-spin prediction is significantly better than a random-label baseline at the same β.

(R2) **Scale-transfer hypothesis:** Under a unified multi-scale transfer protocol, a genuinely RG-like model should exhibit a stable relationship between transfer quality and scale distance and should outperform competitive linear baselines.

(R3) **Turbulence extension target:** A neural closure model trained on subgrid stress reproduces Kolmogorov $k^{-5/3}$ inertial-range scaling in the energy spectrum.

A convincing positive claim of RG-like computation would require aligned evidence across these targets under matched protocol and budget. The present revision provides strong same-scale evidence, but only exploratory evidence for R2 and R3.

Revision scope. The present revision separates results by protocol. The primary quantitative claims use only the unified 10-seed benchmark saved in `outputs/rg_bench/cross_scale/`. An earlier fixed-split pilot ablation (including random-label controls and width sweeps) used a different 3-seed protocol; it is retained only as a sanity check and is *not* numerically mixed with the primary benchmark tables. A separate whole-lattice tiled transfer experiment and dedicated sampler diagnostics are reported with their own reduced exploratory budget and interpreted accordingly.

What this paper establishes, and what it does not: This paper locates itself as a scoped negative-result paper. The strongest supported claim is the same-scale comparison at $L = 16$, β_c: the linear baseline outperforms the MLP in mean test MSE (0.17 ± 0.09 vs. 0.62 ± 0.65), although the strength of this difference is test-sensitive because of the MLP's large variance. The true whole-lattice tiled transfer diagnostic does not support the original monotone-degradation R2 hypothesis, but the current reduced-budget results are better interpreted as a warning that R2 is not yet a stable operational discriminator in this local-rule setting. Combined with the observed sampler autocorrelation at criticality, we treat cross-scale conclusions as exploratory and unresolved. We do not make a primary

quantitative R1 claim from the separate random-label pilot. R3 (turbulence) remains a feasibility demonstration only and does not count as evidence for any criterion.

The main negative finding is: a linear model substantially outperforms the MLP on the same block-spin task, consistent with the majority-vote transformation being dominated by a local linear-threshold rule. This is a precise negative result scoped specifically to *locally linear block-spin tasks*. It does not imply that standard deep architectures cannot learn RG-like computations in general; that would require testing on tasks where the RG transformation is genuinely non-linear in the couplings (e.g., non-zero external field, frustrated magnets, or quantum systems). It also does not contradict the RBM-based positive findings of Mehta & Schwab (2014), which rely on an architectural inductive bias — the bipartite energy function $E(v, h) = -b^\top v - c^\top h - v^\top W h$ — that is absent from standard MLPs. The scope of our claim is limited: standard MLPs do not exhibit RG-like advantages on majority-vote block-spin tasks that are well-approximated by linear projections.

0.2 Background: Ising Model and Block-Spin RG

The 2D Ising model on an $L \times L$ square lattice with periodic boundary conditions has Hamiltonian:

$$H = -J \sum_{\langle ij \rangle} S_i S_j - h \sum_i S_i, \qquad S_i \in \{-1, +1\}. \tag{1}$$

We set $J = 1$, $h = 0$ (zero field). The critical inverse temperature is Onsager's exact value [29]:

$$\beta_c = \frac{1}{2} \log(1 + \sqrt{2}) \approx 0.4407. \tag{2}$$

The 2D Ising model holds a special place in the RG literature for three interrelated reasons. First, Lars Onsager's 1944 exact solution is one of the very few non-trivial phase transitions in statistical mechanics that can be solved analytically in closed form. This provides *ground truth* for all relevant quantities: the free energy, correlation functions, and critically, the critical exponents $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$. Any numerical method for studying RG must be tested against these exact values. Second, the 2D Ising model is the *simplest* model with a non-trivial continuous phase transition: it has a paramagnetic phase at high temperature, a ferromagnetic phase at low temperature, and a second-order transition at β_c with well-defined critical exponents. It captures the essential physics of the ferromagnetic-paramagnetic transition without the complications of additional order parameters, anisotropy, or frustration. Third, the 2D Ising universality class is remarkably robust: any spin model with short-range interactions and a discrete symmetry (Z_2 inversion) on a 2D lattice belongs to the same universality class and shares the same critical exponents. This universality means the 2D Ising results generalize broadly, making it an ideal testbed for controlled theoretical experiments [12, 1, 5].

The block-spin RG transformation with block size $\Lambda = 2$ is:

$$S'_{\lfloor i/\Lambda \rfloor, \lfloor j/\Lambda \rfloor} = \mathrm{sgn}\Big(\sum_{k,l \in \mathrm{block}(i,j)} S_{k,l} \Big), \tag{3}$$

i.e., majority vote within each 2×2 block. Iterating Eq (3) generates an RG flow. At β_c the flow converges to the Wilson-Fisher fixed point; away from β_c it flows to ordered or disordered phases.

The majority-vote construction deserves careful explanation. The Ising order parameter is the *magnetisation* $m = \langle S_i \rangle$, a sum over spins that is discontinuous (non-analytic) at β_c in the thermodynamic limit. At the critical point, the block-averaged spin $\sum_{k,l \in \text{block}} S_{k,l}$ is a continuous random variable (it takes values in $\{-4, -2, 0, 2, 4\}$), but the magnetisation of the coarse-grained system is the sign of this sum. The sign function is a non-linear threshold that implements the discontinuous phase transition: below β_c the sum is overwhelmingly positive (ordered phase, all spins aligned), while above β_c the sum fluctuates around zero (disordered phase). At β_c itself, the sum can be positive or negative with roughly equal probability, and the sign function determines whether the coarse-grained block falls in the up or down domain. Geometrically, the sign function partitions the 4-dimensional spin space (± 1 per spin) into two hemispheres separated by the plane $\sum_i S_i = 0$. This is a *linear* threshold in the space of block-summed features, which explains why the majority-vote transformation is well-approximated by a linear model — the non-linearity enters only through the sign function applied to a single linear combination of the four spins in each block.

Critical exponents for the 2D Ising universality class are exactly known: $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$ [29, 19, 36].

Each critical exponent governs a specific physical quantity near β_c. The exponent ν controls how fast the correlation length ξ diverges as the critical point is approached from either side: $\xi \sim |\beta - \beta_c|^{-\nu}$. Physically, near β_c fluctuations become large and long-range; $\nu = 1$ in 2D Ising means this divergence is relatively slow. The exponent η characterises the decay of the spin-spin correlation function at the critical point itself: $G(r) \sim r^{-(d-2+\eta)}$ with $\eta = 1/4$. In 2D this gives $G(r) \sim r^{-1.75}$, a power law slower than the mean-field prediction r^{-1} because strong fluctuations in 2D modify the correlation structure. The exponent β governs the rise of the order parameter (magnetisation) below β_c: $m \sim (-\epsilon)^\beta$ where $\epsilon = (\beta_c - \beta)/\beta_c$ is the reduced temperature. The value $\beta = 1/8$ (much smaller than the mean-field value $\beta = 1/2$) reflects the difficulty of ordering a 2D system against thermal fluctuations. These three exponents are not independent; they satisfy Rushbrooke's relation $2 - \alpha = d\nu$, Josephson scaling $\nu d = 2 - \eta + \beta(\delta - 1)$ and others, forming a tightly constrained set that the Onsager solution uniquely determines [12, 1].

0.3 Neural Block-Spin Learning Framework

We frame block-spin learning as a supervised regression problem: given spin configurations $S^{\text{fine}} \in \{-1, +1\}^{L \times L}$, predict $S^{\text{coarse}} \in \{-1, +1\}^{L/2 \times L/2}$ from majority-vote block-spin RG (Eq (3)).

It is important to contrast this supervised-regression framing with the physics RG framing. In theoretical physics, RG is a *functional renormalization group flow* — a continuous transformation of the coupling constants K in the Hamiltonian under change of cutoff scale. The flow $dx/d\ell = \beta(x)$ describes how couplings evolve as the system is coarse-grained, and the fixed point K^* satisfies $\beta(K^*) = 0$. This is a continuous, deterministic dynamical system on the space of coupling constants [42, 43, 12, 46]. Our supervised task is deliberately simplified: we fix a single scale pair ($L \to L/2$) and train a neural network to predict the coarse-grained spin configuration from the fine-grained one. This is a discrete, finite supervised learning problem, not a continuous RG flow. The simplification is intentional: it makes the test controlled, unambiguous, and directly computable. If a network cannot learn a single step of block-spin coarse-graining, it is

unlikely to implement the full iterative RG flow. Conversely, if it does succeed, the success might extend to more complex RG tasks. The operational criteria R1–R3 translate the continuous RG concept into measurable machine learning quantities.

Network Architecture

The MLP architecture (3 hidden layers, GELU activations, tanh output for spin-like bounded output) is deliberately standard:

$$\text{MLP} : \mathbb{R}^{L^2} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{128} \xrightarrow{\text{GELU}} \mathbb{R}^{64} \xrightarrow{\tanh} \mathbb{R}^{(L/2)^2}. \tag{4}$$

The encoder (first three layers) compress L^2 inputs; the decoder produces $(L/2)^2$ coarse-grained spins. No explicit RG structure is hardwired. Training: Adam optimizer, lr=10^{-3}, MSE loss, batch size 32, 200 epochs. The unified benchmark reported in the revised tables uses seeds $\{42, \ldots, 51\}$ ($n = 10$). A separate fixed-split pilot ablation uses 3 seeds and is identified explicitly when mentioned. Statistical inference for the primary benchmark uses the saved summaries in `cross_scale_statistics.json`: bootstrap 95% CIs for per-model mean MSE, plus Welch tests and Cohen's d for MLP-vs-Linear comparisons at fixed (L, β).

The choice of this specific MLP architecture deserves justification. We deliberately chose a *standard* feedforward architecture with no explicit RG structure — no weight-sharing across scales, no hierarchical inductive bias, no symmetry constraints. The GELU activation [15] is a smooth approximation to the ReLU that is standard in modern deep learning. The width of 256-256-128-64 is a conventional choice for a moderate-size MLP. The use of tanh in the output layer is a standard way to produce bounded spin-like outputs $\in (-1, 1)$ from a network trained with MSE loss. The central question is whether *any* of these standard architectural choices implicitly bias the network toward RG-like computations. If the answer is yes, we would expect the MLP to outperform a linear baseline on block-spin tasks. If the answer is no, then RG-like inductive biases must be explicitly engineered (as in Neural RG papers), not discovered through standard architecture design.

The linear baseline serves as the minimal model for this task. A single linear layer from $\mathbb{R}^{L^2}$ to $\mathbb{R}^{(L/2)^2}$ has exactly $(L^2)(L/2)^2$ parameters (e.g., for $L = 16$: $256 \times 64 = 16{,}384$ weights). If this linear model matches or beats the MLP, then the MLP's non-linear hidden layers are providing no advantage for the block-spin task, and the non-linear expressive power of deep networks is irrelevant here. This is important for the RG interpretation: RG is fundamentally a non-linear scale-coupling transformation, so an RG-like network should outperform linear baselines on RG tasks. Finding the opposite confirms that the block-spin transformation at $L = 16$ is approximately linear in the spin variables.

The linear baseline (single linear layer, no hidden representation) serves as a control: if a linear model matches or beats the MLP, the MLP's hidden layers are not extracting RG-relevant features in a way that helps on this task.

Heuristic: Majority Vote as a Local Linear-Threshold Map

Let $T : \{-1, +1\}^{L \times L} \to \{-1, +1\}^{L/2 \times L/2}$ denote the exact block-spin RG transformation:

$$T(s)_{IJ} \;=\; \text{sign}\Big(\sum_{i \in B_{IJ}} s_i\Big), \qquad B_{IJ} = \{(2I{-}1, 2J{-}1), \ldots, (2I, 2J)\}, \tag{5}$$

and let

$$z_{IJ}(s) \;=\; \sum_{i \in B_{IJ}} s_i \in \{-4, -2, 0, 2, 4\} \tag{6}$$

be the block sum. Each coarse spin is therefore produced by applying a threshold to a *linear* score on a local 2×2 block:

$$T(s)_{IJ} = \operatorname{sign}\Big(z_{IJ}(s)\Big). \tag{7}$$

This observation is only a heuristic. It does *not* imply a uniform approximation theorem in L or β, and we make no such claim in this revision. The point is more modest: a shallow model can compute each block sum exactly with a sparse linear map, leaving only the final threshold non-linearity. That local linear-threshold structure makes it plausible that a linear baseline can be competitive on the supervised block-spin task. Whether that heuristic predicts actual test-time performance remains an empirical question, answered by the benchmark in Section 0.4.

0.4 Experiments: 2D Ising Model

All experiments use single-spin Metropolis-Hastings Monte Carlo with $N_{\text{train}} = 500$, $N_{\text{test}} = 300$, equilibration 10^3 steps [27, 14].

Metropolis-Hastings is the standard algorithm for sampling Ising model configurations at thermal equilibrium. Starting from an initial spin configuration, the algorithm proposes flipping a single spin, accepts the flip with probability $\min(1, \exp(-\Delta E/T))$ where ΔE is the energy change, and rejects it otherwise. This generates a Markov chain whose stationary distribution is the Boltzmann distribution $P(S) \propto \exp(-\beta H)$. The equilibration period of 10^3 steps is a pragmatic burn-in choice rather than a proof of mixing. In the current scripts we record successive post-burn-in states without an explicit integrated-autocorrelation-time estimate or thinning interval. At β_c, critical slowing down implies that within-chain samples remain correlated, so we do *not* claim that the resulting configurations are independent draws from the thermodynamic ensemble. Separate train/test chains reduce direct data reuse between splits, but the reported error bars should be interpreted as benchmark variability under this sampler budget, not as a certified independent-sample uncertainty quantification. A dedicated diagnostic run reported in Appendix .1 gives $\tau_{\text{int}}(E) = 6.9 \pm 1.9$ and $\tau_{\text{int}}(|M|) = 9.6 \pm 3.0$ sweeps at $(L, \beta) = (8, \beta_c)$, and 18.8 ± 7.3 and 37.0 ± 15.1 sweeps at $(16, \beta_c)$, whereas away from criticality the corresponding values remain near 2–3 sweeps. This quantifies the critical slowing-down caveat rather than leaving it qualitative. Cluster algorithms such as Swendsen-Wang and Wolff would generally be better suited to this regime, but they are not used in the current benchmark implementation [37, 44, 35, 28, 4]. The appropriate interpretation of the reported error bars and significance tests is therefore that of a finite-budget benchmark under correlated sampling, rather than thermodynamic-ensemble-level inference under effectively independent draws. At β_c, the corresponding effective sample sizes are only approximately 218 and 156 for $(L, \beta) = (8, \beta_c)$, and 80 and 41 for $(16, \beta_c)$, for E and $|M|$ respectively.

Evidence tiers. To keep evidence levels explicit, Table 0.1 summarizes the three protocol tiers used in this revision.

Table 0.1: Evidence tiers used in the revised manuscript.

Tier	**Protocol**	**Budget**	**Role**
Main benchmark	Same-scale Ising benchmark	$n = 10$, 500/300, 200 epochs	Primary evidence
Exploratory transfer	Whole-lattice tiled transfer	$n = 3$, 200/100, 40 epochs	Methodological clarification
Appendix diagnostics	Pilot / turbulence / mixing	mixed auxiliary budgets	Non-primary support

Result 1: Same-Scale Block-Spin Benchmark

Setup. The primary quantitative claims in this revision use the unified 10-seed benchmark saved in `outputs/rg_bench/cross_scale/`. For each $L \in \{4, 8, 16\}$, $\beta \in \{0.30, \beta_c, 0.60\}$, and model in {MLP, Linear, CNN, RGMLP}, we train with the same optimizer and data budget and report held-out test MSE. Earlier fixed-split pilot ablations (random-label controls and width sweeps) were generated by a different 3-seed protocol and are therefore *not* mixed numerically with the tables below.

Results. Table 0.2 and Figure 0.1.

Table 0.2: Main same-scale Ising benchmark ($n = 10$ seeds). All models are trained on block-spin labels under the unified protocol saved in `cross_scale_statistics.json`. Values are mean ± std dev over seeds $\{42, \ldots, 51\}$. This is the main benchmark of record and is not directly comparable in evidence strength to the reduced-budget exploratory transfer table below.

Model	β	$L = 4$ **MSE**	$L = 8$ **MSE**	$L = 16$ **MSE (same-L)**
MLP	0.30	0.0005 ± 0.0006	0.1198 ± 0.0128	0.6006 ± 0.0287
MLP	β_c	0.0307 ± 0.0239	0.2017 ± 0.2187	0.6249 ± 0.6456
MLP	0.60	0.0168 ± 0.0231	0.1973 ± 0.5341	0.1178 ± 0.1695
Linear	0.30	0.0314 ± 0.0059	0.0536 ± 0.0051	0.4160 ± 0.0123
Linear	β_c	0.0357 ± 0.0265	0.0692 ± 0.0309	0.1736 ± 0.0862
Linear	0.60	0.0159 ± 0.0216	0.0140 ± 0.0092	0.0106 ± 0.0099
CNN	0.30	$3.2 \times 10^{-5} \pm 0.9 \times 10^{-5}$	$1.0 \times 10^{-5} \pm 0.1 \times 10^{-5}$	$3.6 \times 10^{-6} \pm 0.5 \times 10^{-6}$
CNN	β_c	$5.4 \times 10^{-4} \pm 1.1 \times 10^{-3}$	$2.8 \times 10^{-5} \pm 2.9 \times 10^{-5}$	$6.0 \times 10^{-6} \pm 5.0 \times 10^{-6}$
CNN	0.60	0.0248 ± 0.0432	0.0451 ± 0.0714	0.0028 ± 0.0073
RGMLP	0.30	0.4928 ± 0.3231	0.7612 ± 0.3163	1.1784 ± 0.2744
RGMLP	β_c	0.1542 ± 0.0672	0.5402 ± 0.3147	1.0646 ± 1.0284
RGMLP	0.60	0.6601 ± 1.3218	1.7935 ± 1.4374	0.9792 ± 1.2162

Key findings from Table 0.2:

(i) **At $L = 16$ and β_c, the linear baseline beats the MLP.** The unified same-scale benchmark gives 0.1736 ± 0.0862 for Linear vs. 0.6249 ± 0.6456 for MLP at $(L, \beta) = (16, \beta_c)$. The mean ordering favors the linear model, but the strength of the difference depends on the test because the MLP variance is large (Welch $p = 0.055$, Mann-Whitney $p = 0.026$, Cohen's $d = -0.98$). The critical-point task is therefore learnable, but the extra non-linear capacity of the MLP does not provide a robust same-scale advantage.

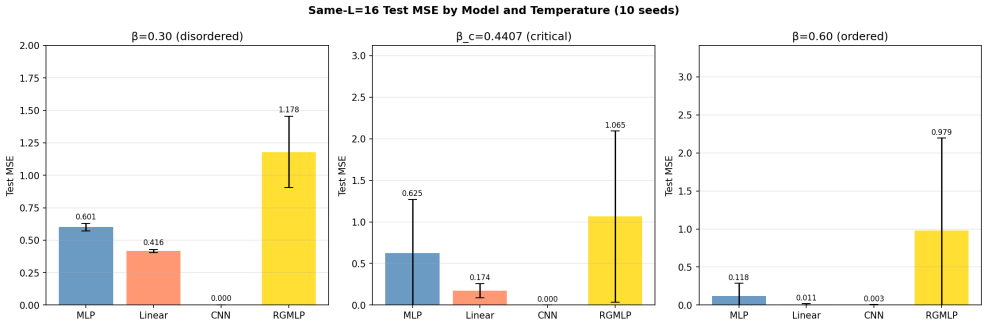

Figure 0.1: Within-scale Ising block-spin performance across β. The CNN reaches near-zero MSE in most tested settings, with visible degradation only in the low-L ordered cases at $\beta = 0.60$, while the linear model matches or outperforms the MLP at the critical point.

(ii) **Error depends strongly on phase and lattice size.** At $L = 16$, the linear model is easiest in the ordered phase (0.0106 ± 0.0099 at $\beta = 0.60$), while the MLP remains high in both the disordered and critical regimes (0.6006 ± 0.0287 at $\beta = 0.30$ and 0.6249 ± 0.6456 at β_c). The temperature dependence in Figure 0.2 shows that the critical regime carries the largest seed-to-seed instability. In particular, the disordered same-scale setting $(L, \beta) = (16, 0.30)$ yields an extreme effect size ($d = -8.36$) because the mean gap is large while both models have small standard deviation.

(iii) **Architectural bias matters more than generic depth.** The CNN with explicit locality and translation structure reaches near-zero MSE across nearly all (L, β) settings, whereas RGMLP is unstable and often worse than the plain MLP. The task is therefore perfectly learnable with an appropriate inductive bias, but standard fully connected depth alone is not enough. This is not, by itself, evidence that the CNN has learned a physically meaningful RG flow; it shows only that locality and translation structure are sufficient for the supervised block-spin map.

The earlier random-label control is retained only as an appendix-level pilot sanity check and is not used for the main quantitative claims in this revision.

Figure 0.2 summarizes the within-scale temperature trend at $L = 16$: both models improve away from criticality, and the ordered phase yields the lowest MSE for both models.

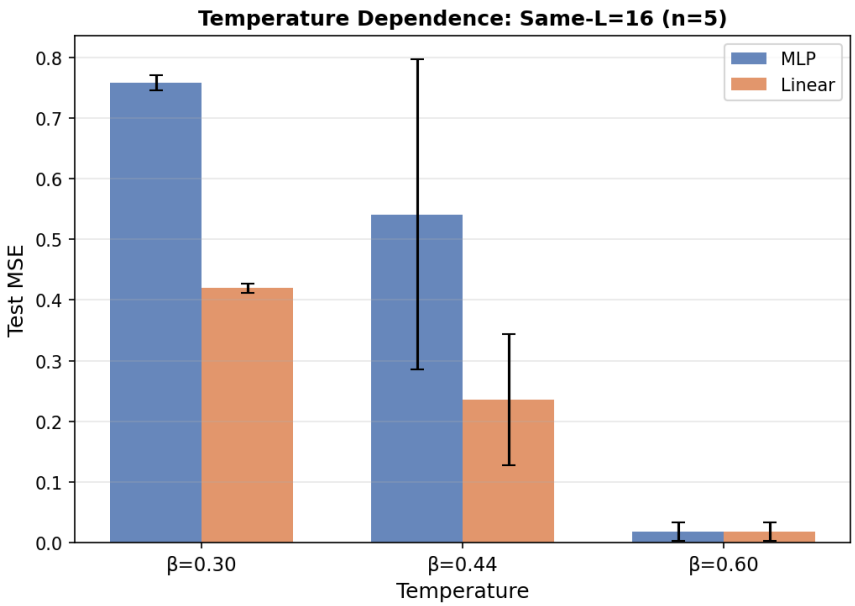

Figure 0.2: Temperature dependence of within-scale block-spin MSE at $L = 16$. Both models improve away from criticality, and the ordered phase yields the lowest MSE for both models.

Result 2: Whole-Lattice Tiled Transfer (Exploratory)

Setup. To replace the earlier patch-only proxy with a true whole-lattice evaluation, we trained source-scale predictors on $L \in \{4, 8\}$ and then tiled the learned local model over the full target lattice. Concretely, a model trained on $L = 8$ receives each non-overlapping 8×8 patch of a 16×16 configuration, predicts the corresponding 4×4 coarse block, and the four predicted blocks are assembled into the full 8×8 coarse lattice. The same construction yields $4 \to 8$ and $4 \to 16$ transfer. Because this experiment was added after the main 10-seed benchmark, we ran it under a reduced exploratory budget ($n = 3$ seeds, $N_{\text{train}} = 200$, $N_{\text{test}} = 100$, 40 epochs) and compare transfer against same-budget same-L baselines from the same script.

Table 0.3: Exploratory whole-lattice tiled transfer at β_c. Same-budget baselines and transfers come from the `whole_lattice_transfer/summary.json` using $n = 3$ seeds, $N_{\text{train}} = 200$, $N_{\text{test}} = 100$, and 40 epochs. "Deg." is relative to the same-budget same-$L = 16$ baseline for the same model.

Source → 16	**MLP**	**MLP Deg.**	**Linear**	**Linear Deg.**
$16 \to 16$	0.463 ± 0.174	$1.00\times$	0.287 ± 0.170	$1.00\times$
$8 \to 16$	0.293 ± 0.190	$0.63\times$	0.244 ± 0.129	$0.85\times$
$4 \to 16$	0.141 ± 0.107	$0.31\times$	0.222 ± 0.113	$0.77\times$

- **The true tiled experiment does not support monotone degradation.** At β_c, neither model deteriorates when transferred from $L = 8$ to $L = 16$ under this tiled evaluation, and the $4 \to 16$ transfer is numerically *better* still. The originally proposed monotone scale-distance interpretation therefore fails in this local-rule setting.

- **Linear baselines remain fully competitive.** Although the MLP improves substantially relative to its same-budget $16 \to 16$ baseline, the linear model also transfers competitively: 0.244 ± 0.129 for $8 \to 16$ and 0.222 ± 0.113 for $4 \to 16$, versus 0.287 ± 0.170 at $16 \to 16$.

- **The cross-scale story is protocol-sensitive.** The new whole-lattice tiled evaluation gives a different qualitative picture from the earlier patch-only proxy, which means cross-scale conclusions depend strongly on how transfer is operationalized. Appendix .2 (Table .5) shows that this non-monotone behavior is not confined to β_c.

R2 assessment. The new whole-lattice experiment removes the "top-left patch" ambiguity of the earlier proxy, but it still does not deliver a stable positive RG signature. Instead, it shows that transfer in this benchmark is highly protocol- and budget-sensitive: the tiled whole-lattice evaluation does *not* display monotone degradation with scale distance, and linear baselines remain competitive throughout. The right interpretation is therefore not that R2 has been cleanly falsified, but that for the present local-rule setting the original R2 formulation is not yet a stable operational discriminator. We therefore keep R2 unresolved.

Auxiliary analyses moved to the appendix. Legacy pilot sanity checks, sampler diagnostics, and the turbulence feasibility experiment are reported in Appendices .1–.4 and are not used as primary main-text evidence.

0.5 Discussion

Why Does the Linear Model Outperform the MLP?

The strongest supported result in the revised manuscript is the unified same-scale benchmark at $(L, \beta) = (16, \beta_c)$: Linear reaches 0.1736 ± 0.0862 while MLP reaches 0.6249 ± 0.6456 ($n = 10$). We focus on this result because it comes from the same 10-seed protocol used throughout the main tables; the earlier fixed-split random-label pilot is not used for the primary quantitative claim.

We offer three non-mutually-exclusive interpretations:

(i) **Target simplicity near β_c:** The block-spin majority-vote at criticality is effectively a *linear* threshold function: each coarse spin is the sign of a local sum of four ± 1 inputs. A shallow linear map can compute those block sums exactly, leaving only the final threshold non-linearity. This local linear-threshold structure makes it plausible that extra hidden-layer non-linearities are unnecessary for the supervised task studied here.

(ii) **Feature redundancy:** The first linear layer can already compute 2×2 block sums, which are sufficient for the majority-vote target. Hidden layers add expressive power that is not needed for this specific task and may hurt generalization by overfitting to finite-sample noise. In our setting there is no evidence that generic depth discovers a better scale-coupled representation than the linear baseline; the CNN result suggests that architectural bias, not depth alone, is the missing ingredient.

(iii) **Training dynamics:** The MLP may converge to different local minima across seeds, leading to high variance. The unified benchmark records standard deviations of 0.646 for the MLP and 0.086 for the linear model at $(L, \beta) = (16, \beta_c)$. This instability is itself part of the negative result: the MLP's added flexibility manifests as optimization variance rather than as a reliable accuracy advantage.

This finding suggests that, on the present benchmark, the RG-relevant computation is dominated by a local linear-threshold rule rather than by a hierarchical non-linear representation.

Implications for RG-like Behavior Assessment

The whole-lattice tiled experiment sharpens the methodological lesson. Once transfer is evaluated on the full target lattice rather than on a single patch, the original monotone-degradation criterion still does not emerge. At β_c, both $8 \to 16$ and $4 \to 16$ transfer remain competitive with the same-budget same-$L = 16$ baseline, and the smaller source scale can even perform better numerically. This means that, in the present local-rule setting, R2 is not a stable discriminator of RG-like behavior.

Three non-exclusive mechanisms may explain why smaller source scales can appear to transfer better in the tiled experiment. First, *local-rule tileability*: once the source model

has learned a sufficiently accurate local majority-vote rule, that rule can be repeated across the larger lattice without needing a new global representation. Second, *source-task simplification*: the 4×4 and 8×8 source problems may be easier optimization problems than the 16×16 one, so a smaller source scale can yield a cleaner local predictor. Third, *budget mismatch*: under the reduced exploratory budget, the same-$L = 16$ baselines may remain under-optimized relative to the tiled transfer models, which weakens any attempt to read the ratios as intrinsic RG flow behavior. This suggests that tiled transfer may primarily probe reusable locality of the learned block rule, rather than hierarchical scale-coupling in the RG sense.

The across-temperature whole-lattice results strengthen this interpretation: non-monotone transfer is not a critical-point-specific anomaly, but looks more like a protocol artifact of tiled local-rule reuse across source scales.

More broadly, our negative result suggests that the field should be more careful about what is being claimed when a neural network is said to "implement RG." If a task is simple enough to be solved by a linear baseline, then apparent RG-like structure in the learned representation may be coincidental rather than causal. Stronger claims should be reserved for settings in which linear baselines fail, transfer protocols match the stated operational criterion, and sampling uncertainty is characterized more carefully than in the present benchmark.

0.6 Related Work

There are now two largely separate literatures around machine learning and statistical mechanics. One uses supervised or unsupervised learning to detect phases, order parameters, and transition points in classical or quantum lattice systems [39, 7, 38, 17, 40, 8, 3, 45, 33, 24]. More interpretable variants ask what features the model has actually learned, for example through kernel methods, explicit order parameters, or symmetry diagnostics [31, 41, 13, 23, 9]. Our benchmark is narrower than that literature: we do not ask whether a network can detect a phase transition, but whether it gains a specifically RG-like advantage on a known coarse-graining map.

A second literature aims more directly at RG itself. Mehta and Schwab [26] connected RBMs with variational RG; Koch-Janusz and Ringel [20] recast coarse graining in terms of mutual information; Li and Wang [22], Pang et al. [30], Hu et al. [16], and Bachtis et al. [2] proposed architectures or objectives that encode multi-scale structure explicitly. Those papers are valuable precisely because their model classes are RG-motivated. Our question is stricter and more negative-result oriented: if one removes the explicit RG inductive bias and compares against strong linear baselines, does a standard fully connected MLP still exhibit an RG-like advantage on the supervised block-spin task?

Finally, our treatment of sampling is informed by the Monte Carlo literature. Local Metropolis updates [27, 14] are known to slow dramatically near criticality, while cluster methods such as Swendsen-Wang and Wolff mitigate this problem [37, 44, 35, 28, 4]. General reviews of machine learning in physics provide the broader context for why careful benchmark design matters here too [6, 25]. Relative to that prior work, the main contribution of the present paper is methodological: a matched-budget benchmark that keeps same-scale evidence, exploratory transfer, and sampler diagnostics explicitly separated, and that shows how easily broad RG language can outrun what a local majority-vote task actually supports.

0.7 Conclusion

This revision supports a scoped negative result. In the unified 10-seed same-scale benchmark, the linear baseline has lower mean test MSE than the MLP at the critical point: at $(L, \beta) = (16, \beta_c)$ the MLP reaches 0.6249 ± 0.6456, while Linear reaches 0.1736 ± 0.0862. The mean ordering favors the linear model, but the strength of the difference depends on the test because the MLP variance is large (Welch $p = 0.055$, Mann-Whitney $p = 0.026$, Cohen's $d = -0.98$). Across the benchmark as a whole, no setting shows a robust MLP advantage over the linear baseline. The supported claim is therefore specific: on majority-vote block-spin tasks dominated by a local linear-threshold structure, standard fully connected depth does not provide an RG-like advantage over simpler linear projections.

The exploratory whole-lattice tiled transfer and the sampler diagnostics mainly clarify methodology rather than add parallel primary evidence. At β_c, the tiled transfer protocol does not exhibit monotone degradation with scale distance: $8 \to 16$ and even $4 \to 16$ remain competitive with the same-budget $16 \to 16$ baseline, and linear models stay competitive throughout. At the same time, the critical-point sampler diagnostics show substantial autocorrelation, with $\tau_{\text{int}}(E) = 18.8 \pm 7.3$ and $\tau_{\text{int}}(|M|) = 37.0 \pm 15.1$ sweeps at $(16, \beta_c)$, implying effective sample sizes of only about 80 and 41 for the diagnostic run. We therefore treat R2 as unresolved rather than falsified, and we do not promote the appendix-level turbulence note or the older random-label pilot to primary evidence.

The durable contribution of the paper is thus methodological. A convincing positive claim of RG-like computation should align matched training budgets, true multi-scale chains (for example $4 \to 8 \to 16$ together with direct $4 \to 16$ and $8 \to 16$ transfer), autocorrelation-aware sampling, and a regime in which competitive linear baselines actually fail. All main tables in this revision are sourced from the separated raw outputs in `outputs/rg_bench/cross_scale/`, `outputs/rg_bench/whole_lattice_transfer/`, and `outputs/rg_bench/mixing_analysis/`, with the legacy pilot outputs kept appendix-only. On the present local majority-vote block-spin benchmark, that combination of evidence does not support a claim that standard MLPs learn RG-like advantages.

.1 Sampling Diagnostics

To quantify the sampling caveat rather than leave it qualitative, we ran a dedicated time-series diagnostic using 3 seeds, 3000 post-burn-in sweeps, and the observables E and $|M|$. Table .4 reports the resulting integrated autocorrelation times and mean acceptance rates.

Table .4: Metropolis-Hastings mixing diagnostics from the `mixing_analysis/summary.json` output.

Setting	$\tau_{\text{int}}(E)$	$\tau_{\text{int}}(\lvert M\rvert)$	**Acceptance**
$L = 8$, $\beta = 0.30$	1.7 ± 0.4	2.7 ± 0.4	0.520 ± 0.004
$L = 8$, β_c	6.9 ± 1.9	9.6 ± 3.0	0.171 ± 0.005
$L = 8$, $\beta = 0.60$	1.8 ± 0.3	1.9 ± 0.3	0.0250 ± 0.0006
$L = 16$, $\beta = 0.30$	1.5 ± 0.2	3.1 ± 0.2	0.522 ± 0.001
$L = 16$, β_c	18.8 ± 7.3	37.0 ± 15.1	0.184 ± 0.006
$L = 16$, $\beta = 0.60$	1.9 ± 0.2	2.3 ± 0.3	0.0259 ± 0.0008

The critical-point autocorrelation times are an order of magnitude larger than their off-critical counterparts, especially for $|M|$ at $L = 16$. This confirms that the benchmark should be read as a finite-budget computational study, not as an effectively independent-sample Monte Carlo analysis. With 3000 post-burn-in sweeps, the corresponding effective sample sizes are approximately $3000/(2\tau_{\text{int}}) \approx 218$ for E and 156 for $|M|$ at $(L, \beta) = (8, \beta_c)$, and approximately 80 and 41 at $(16, \beta_c)$.

.2 Whole-Lattice Transfer Across Temperatures

Table .5 reports the reduced-budget tiled whole-lattice transfer for all three temperatures. The non-monotone transfer behavior is therefore not confined to β_c, although its interpretation remains exploratory because the full transfer experiment is only available at $n = 3$ seeds and a smaller training budget than the main benchmark.

Table .5: Exploratory whole-lattice tiled transfer across temperatures from `whole_lattice_transfer/summary.json`. All entries use $n = 3$ seeds, $N_{\text{train}} = 200$, $N_{\text{test}} = 100$, and 40 epochs.

β	**Source → Target**	**MLP**	**Linear**	**Comment**
0.30	$4 \to 8$	0.088 ± 0.044	0.562 ± 0.104	MLP transfer better than same-$L = 8$
0.30	$4 \to 16$	0.084 ± 0.037	0.545 ± 0.081	MLP strongly below same-$L = 16$
0.30	$8 \to 16$	0.344 ± 0.015	0.509 ± 0.013	Both beat same-$L = 16$
β_c	$4 \to 8$	0.095 ± 0.057	0.168 ± 0.088	Non-monotone vs. source scale
β_c	$4 \to 16$	0.141 ± 0.107	0.222 ± 0.113	Better than same-$L = 16$ for both
β_c	$8 \to 16$	0.293 ± 0.190	0.244 ± 0.129	Competitive with same-$L = 16$
0.60	$4 \to 8$	0.019 ± 0.009	0.040 ± 0.007	MLP near same-$L = 8$; Linear degrades
0.60	$4 \to 16$	0.016 ± 0.011	0.046 ± 0.012	Strong model asymmetry
0.60	$8 \to 16$	0.016 ± 0.013	0.017 ± 0.012	Linear near same-$L = 16$

.3 Legacy Pilot Sanity Checks

Earlier fixed-split pilot experiments (random-label controls, width sweeps, and the first ablation figures) were generated under a separate 3-seed protocol and now live in `outputs/rg_bench/pilot_ablation/`. They remain useful as sanity checks, but they are excluded from the main quantitative tables because their data generation, seeding, and training budgets do not match the unified same-scale benchmark.

.4 Exploratory Turbulence Note

The turbulence closure experiment is retained only as a compressed appendix note. Using D2Q9 Lattice Boltzmann data at Re= 200, the exploratory spectrum fit gives $\alpha = -1.30 \pm 0.05$ over $k \in [2, 20]$, compared with the Kolmogorov target $-5/3 \approx -1.67$. The gap is substantial, the inertial range is too short at this Reynolds number, and the experiment therefore does not support R3. It remains a feasibility check for future higher-Re studies.

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